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Section = A Gr.

Ques State and prove maximum power transfer theorem

→ According to this theorem the load resistance R_L will receive maximum power from the ~~source~~ source V_{TH} when load resistance R_L become equals to equivalent resistance R_{TH} . $R_L = R_{TH}$

Proof:-

$$I_L = \frac{V_{TH}}{R_{TH} + R_L}$$

for maximum Power at R_L .

$$\begin{aligned} P_L &= I_L^2 R_L \\ &= \left(\frac{V_{TH}}{R_{TH} + R_L} \right)^2 R_L \\ &= \left(\frac{V_{TH}^2}{R_{TH}^2 + R_L^2 + 2R_{TH}R_L} \right) R_L \end{aligned}$$

Now, multiply ~~and divide~~ the ~~by~~ by $1/R_L$ numerator and denominator

$$= \frac{V_{TH}^2}{\left(\frac{R_{TH}^2}{R_L} \right) + \frac{R_L}{R_L} + 2R_{TH}}$$

let $\frac{R_{TH}^2}{R_L} + R_L + 2R_{TH} = x$

$$\frac{dx}{dR_L} = \frac{d}{dR_L} \left(\frac{R_{TH}^2}{R_L} + R_L + 2R_{TH} \right)$$

$$= -\frac{R_{TH}^2}{R_L^2} + 1 + 0$$

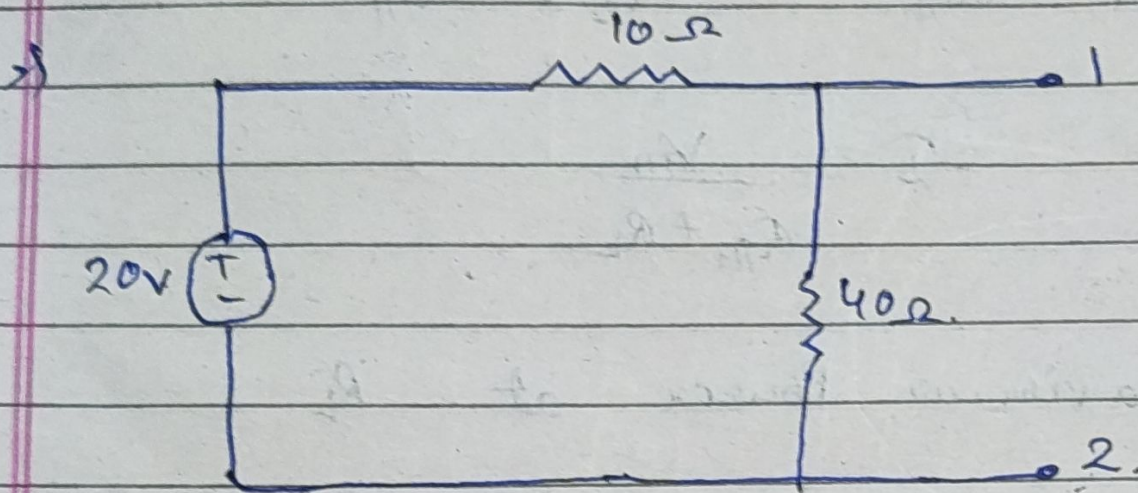
$$\Rightarrow -R_{TH}^2 + R_L^2 = 0 \Rightarrow R_{TH}^2 = R_L^2$$

$$R_{TH} = R_L$$

Hence proved.

Q.2 State Thevenin's theorem.

Ans. Thevenin's theorem states that a complex circuit can be reduced by applying equivalent voltage and load resistance in series.



for V_{TH}

$$20 - 10I - 40I = 0$$

$$20 = 50I$$

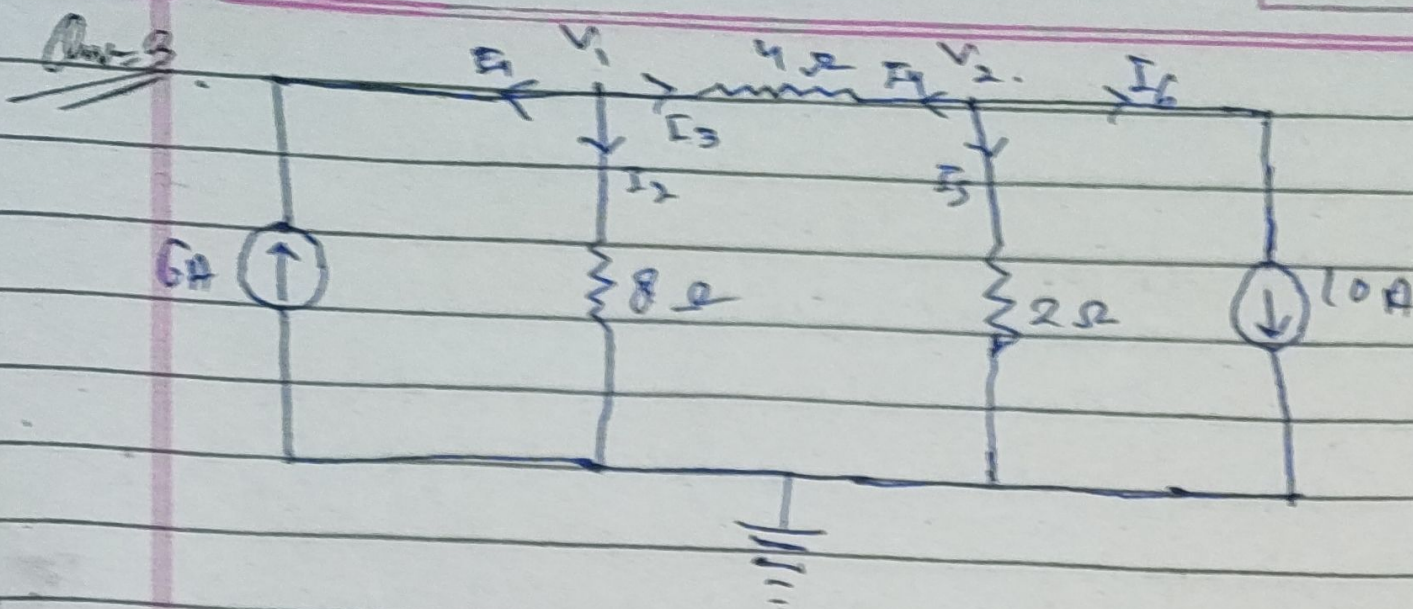
$$I = \frac{20}{50} = 0.4 \text{ A}$$

$$V_{TH} = V_{40} = 40I$$

$$= 40 \times 0.4$$

$$= 16 \text{ V}$$

$$R_{TH} = \frac{10 \times 40}{10 + 40} = \frac{400}{50} = 8 \Omega$$



$$2) \quad I_1 = -6 \text{ A}, \quad I_2 = \frac{V_1}{8}, \quad I_3 = \frac{V_1 - V_2}{4}$$

$$3) \quad I_4 = \frac{V_2 - V_1}{4}, \quad I_5 = \frac{V_2}{2}, \quad I_6 = +10 \text{ A}$$

$\Rightarrow$

$$\begin{aligned} I_1 + I_2 + I_3 &= 0 \\ -6 + \frac{V_1}{8} + \frac{V_1 - V_2}{4} &= 0 \\ \frac{-48 + V_1 + 2V_1 - 2V_2}{8} &= 0 \end{aligned}$$

$$3V_1 - 2V_2 = 48 \quad \text{--- (1)}$$

$$\begin{aligned} 4) \quad I_4 + I_5 + I_6 &= 0 \\ \frac{V_2 - V_1}{4} + \frac{V_2}{2} + 10 &= 0 \\ \frac{V_2 - V_1 + 2V_2 + 40}{4} &= 0 \end{aligned}$$

$$3V_2 - V_1 = -40$$

$$-V_1 + 3V_2 = -40 \quad \text{--- (2)}$$

$$V_1 = 9.14, \quad V_2 = -10.28 \text{ A}$$